

CTO CAPSTONE PROJECT · 2026

CTO360

Enterprise Technology Control Tower

One View. Connected Decisions. Measurable Technology Outcomes.

Moving an enterprise from current state to target state, measurably, with governed AI.



7 source systems · 3 business units · 1 decision layer

Ram
CTO

Suman
Strategy & ROI

Vaibhav
Regulatory & AI Governance

Santhosh
Finance & Investment

Pankaj
Operations

What We Will Cover

About 25 minutes of story and live demo, followed by questions.

01

The Problem

Why CTOs cannot steer AI-led transformation today

02

The Solution

CTO360: a decision layer over existing tools

03

How It Works

Architecture, simulation and how we measure change

04

Live Demo

Five owner workspaces and one connected decision

05

Frameworks & Value

Frameworks applied with metrics, value case, roadmap

06

Q&A

Questions and discussion

A Decision Problem, Not a Data Problem

The CTO cannot see across the enterprise fast enough to decide where to intervene, or prove that technology spend produced business outcomes.

- **Fragmented signals.** Strategy, architecture, process, delivery, operations and risk each live in a separate tool, and each answers only its own question.
- **Cause and effect span tools.** A payments programme slips because of legacy dependencies (LeanIX), a team capacity gap (Jellyfish) and rising incidents (Datadog). No single tool shows that chain, so it is found late.
- **Reporting is not deciding.** Dashboards and status decks are plentiful, but signal to decision to action to outcome is manual, slow and rarely closed.
- **Technology spend is hard to defend.** Value realised and risk avoided are not tied to specific decisions.
- **Every business unit reports differently,** so the CTO cannot compare them or move money and people with confidence.

Planview

answers: Strategy

SAP LeanIX

answers: Architecture

Celonis + Signavio

answers: Process

ServiceNow SPM

answers: Workflow

Jellyfish

answers: Engineering

Datadog

answers: Operations

Vanta

answers: Risk

What matters most?

no tool answers this

Same Problem, Three Different Realities

Three simulated enterprises with different industries, but the same root cause: signals that are never connected into a decision.

BANKING

Meridian Bank

Business problem: Legacy core and integration layer slowing digital payments; technology debt behind P1 incidents.

WHAT IT COSTS

- Q4 payments milestone at risk
- ₹38.9 Cr value at stake
- Regulatory exposure (RBI, DPDP)

MANUFACTURING

Arvant Industries

Business problem: MES and plant system instability causing line stops; smart factory programme delayed.

WHAT IT COSTS

- Lost production hours
- Late deliveries to customers
- OEE target missed

RETAIL

Orbit Retail

Business problem: Festive peak readiness and inventory sync; engineering capacity stretched.

WHAT IT COSTS

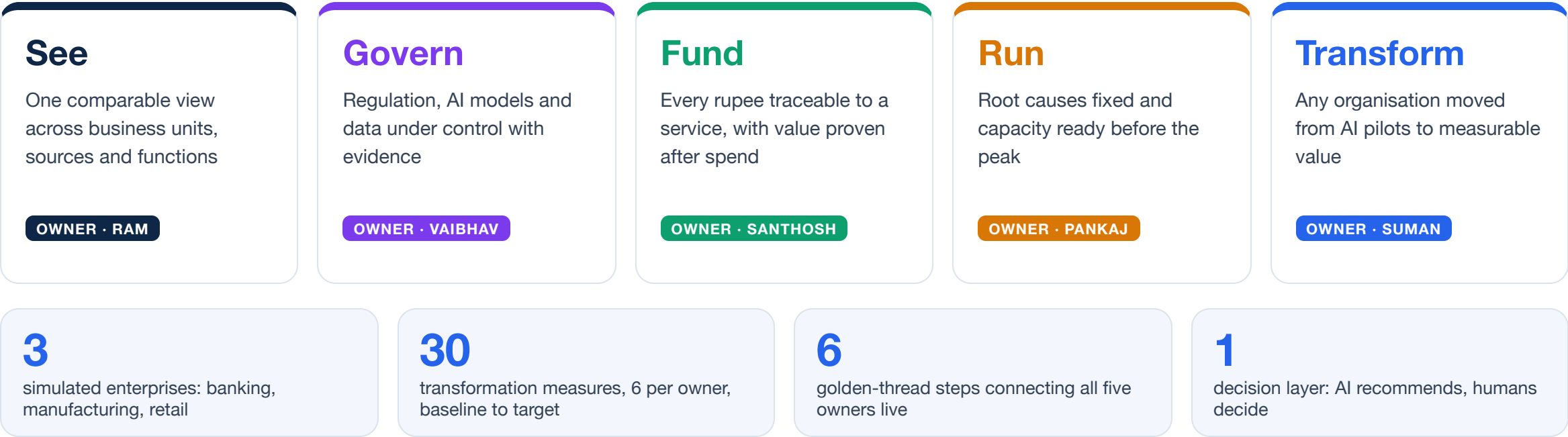
- Lost festive sales
- Overselling and stockouts
- Point-of-sale security risk

The pattern repeats across industries, so the answer must be built once and configured per industry. Retail was added by configuration only.

Our Capstone Problem Statement

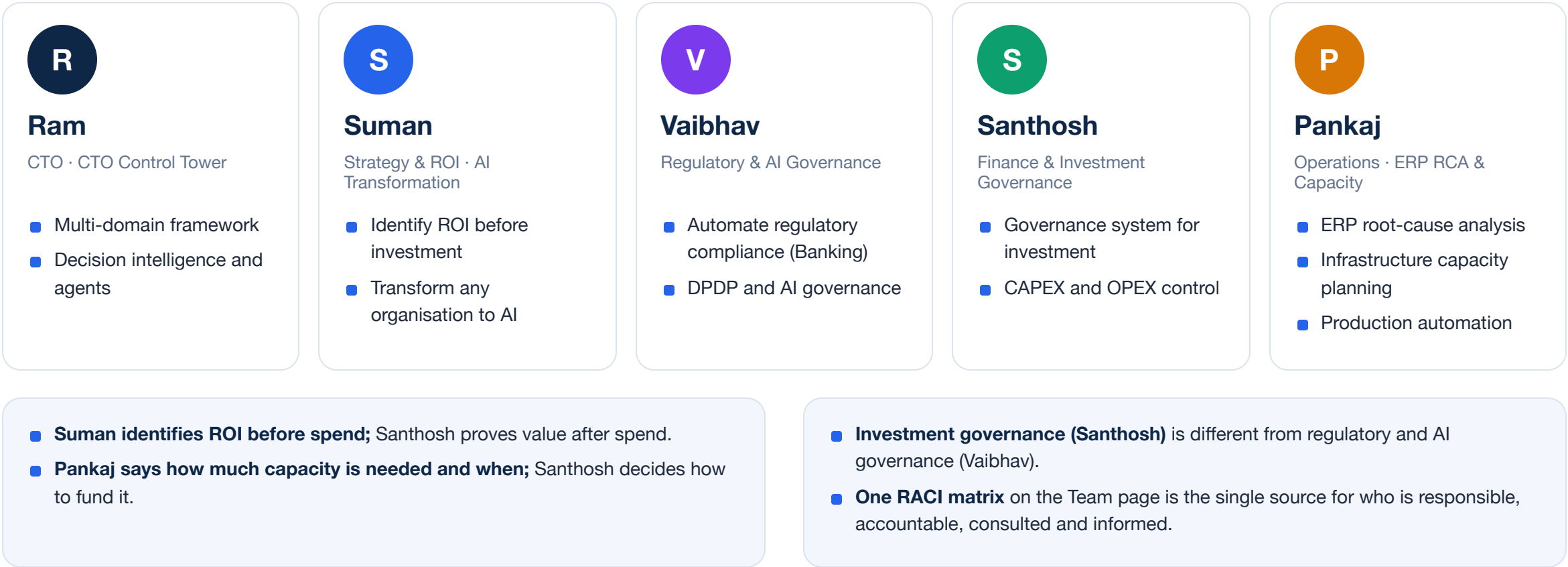
Five verbs, five owners, one system. Each owner takes one real business problem (BP1) from current state to target state.

Enterprises cannot **see, govern, fund and run** AI-led transformation as one system. CTO360 simulates a real enterprise and moves each area from its current state to a target state, **measurably**.



Team, Ownership and Business Problems

Each owner has a dedicated live workspace in CTO360. Clear splits remove overlap between roles.



CTO360: A Decision Layer, Not Another Tool

CTO360 sits on top of the tools an enterprise already owns and turns their signals into decisions that are owned, acted on and measured.



- **One view:** 3 business units × 7 source systems × 16 enterprise functions, all scored the same way so they can be compared.
- **Correlation:** records are linked across tools through shared IDs, so every decision card shows why it exists, with traceable evidence.
- **Prioritised Top 10:** each decision shows impact, money asked versus value protected, confidence, owner and deadline.
- **Closed loop:** approve a decision, issue actions, and track the outcome against baseline and target.

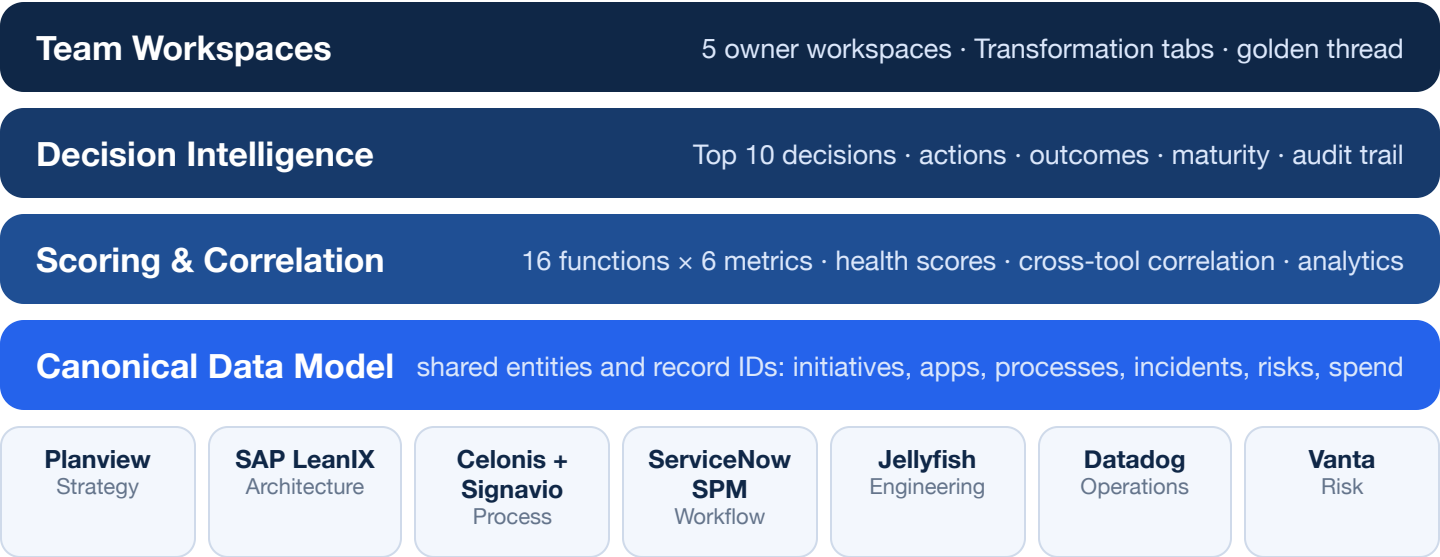
GOVERNANCE · TEDIF

AI Recommends. Humans Decide.

Trusted Enterprise Decision Intelligence Framework: AI gathers signals and recommends, a named person decides at a gate, and every step is recorded in an audit trail.

Architecture: 3 Business Units × 7 Sources × 16 Functions

A layered design: sources at the bottom, a common data model, scoring and correlation, then decisions and team workspaces on top.

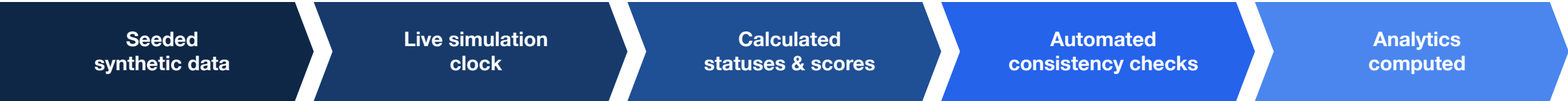


7 simulated source systems (one per enterprise question)

- **Build once, configure per industry:** Meridian Bank, Arvant Industries and Orbit Retail run on the same engine.
- **7 source systems:** each represents a real enterprise tool category and feeds the canonical model.
- **16 enterprise functions:** from Technology and Data & AI to Finance, HR, Legal and Procurement, each with 6 scored metrics.
- **Shared record IDs** are what make cross-tool correlation, evidence and drill-down possible.

A Realistic Simulation, Built to Be Trusted

The simulation behaves like a real enterprise: data, events and calculations are consistent, repeatable and checked automatically.



- **Seeded data:** the same realistic enterprise is generated every time, so the demo is repeatable.
- **Live clock:** the simulation runs in real time; events such as a new regulatory circular or a breach clock arrive on stage.
- **Nothing is typed by hand:** every status, score, risk level and decision priority is calculated from the records.
- **Automated checks:** every build validates 3 domains plus team cross-checks, for example spend ledgers reconcile and every obligation maps to a real control.

Analytics Actually Computed

Holt-Winters forecasting

EWMA anomaly detection

Monte Carlo (P10 / P50 / P90)

NPV · IRR · payback

S-curve adoption

PSI model drift

Pareto 80/20

Queueing (M/M/1)

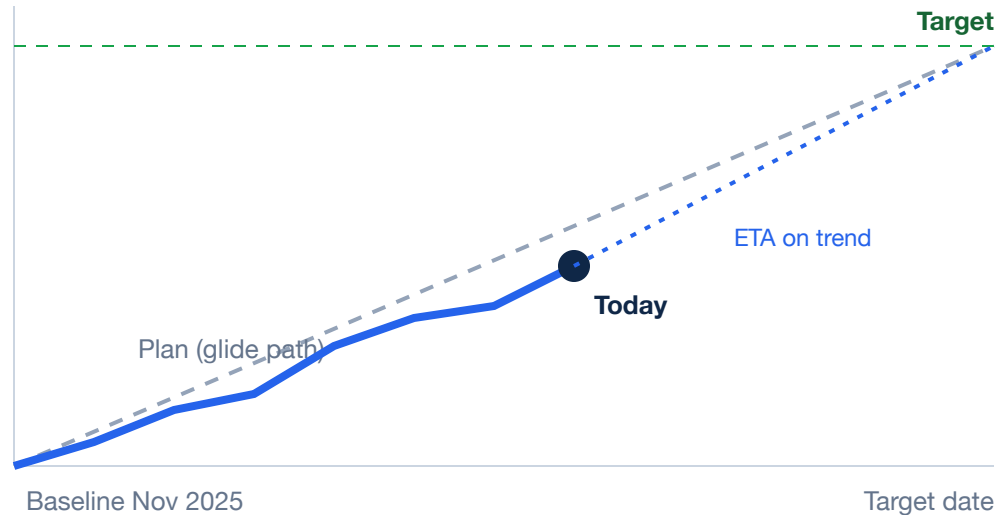
SRE error budget

Glide path & ETA

Honest framing: this is synthetic data, not client data. The design takes real tool exports as the next step.

How We Measure Transformation

Every owner moves from old state to target state. Progress is compared with time used, so we know whether we are transforming fast enough.



- **6 measures per owner, 30 in total**, each with an old state (baseline Nov 2025), today, and a target with a date.
- **Progress** = share of the gap from baseline to target already closed.
- **Time used** = share of the time from baseline to the target date already passed.
- **Glide path**: comparing progress with time used gives Ahead, On Track or Behind, so a slow start is visible early.
- **ETA**: projected from the last quarter's trend; flagged when the trend is flat or moving away.
- **Same rule for everyone**, so a team score (about 59% today) can be compared across owners.

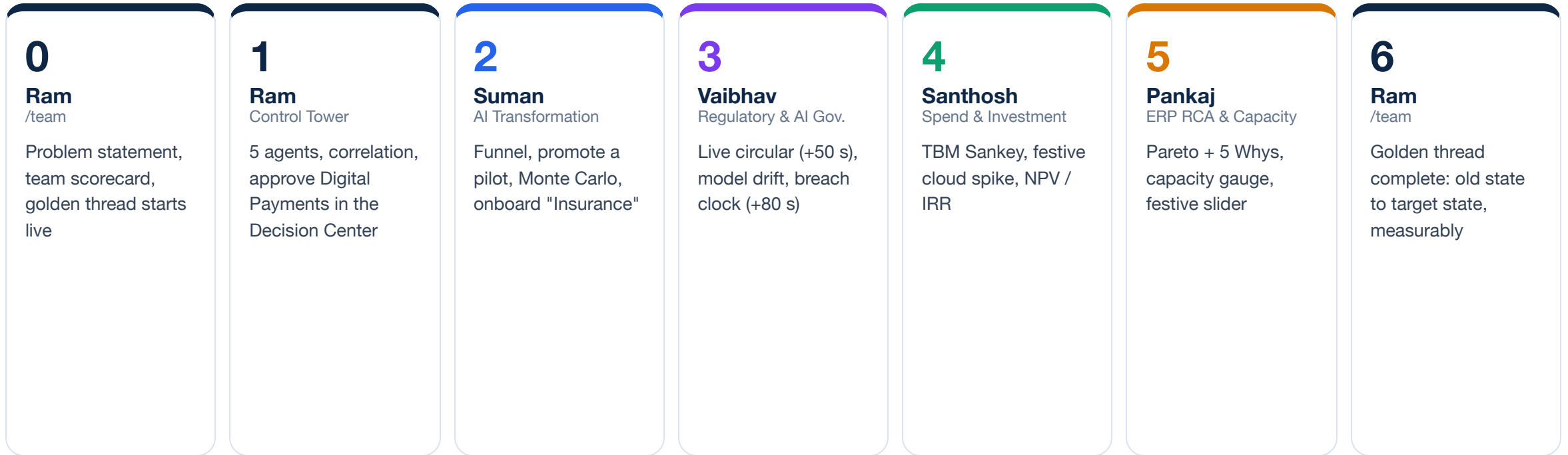
Ahead
progress $\geq$ time used + 10

On Track
within ± 10 points

Behind
progress < time used - 10

Live Demo: Five Owners, One Connected Decision

Live site: cto360.vercel.app. The scenario plays on a simulation clock while we present.



- **Before starting:** open the Team page and press Reset in the header so the scenario replays from the beginning.
- **Each presenter, about 3 minutes:** Transformation tab, two functional tabs with one live action, then the Frameworks Applied tab.

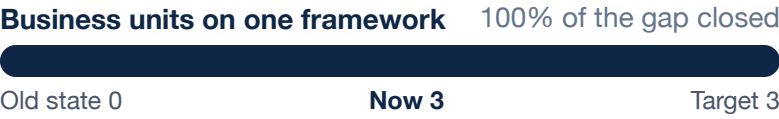
Ram: CTO Control Tower

INTEGRATE, DECIDE, PROVE OUTCOMES **BP1:** Multi-domain framework; decision intelligence and agents

OLD STATE
Signals scattered across 7 tools and 3 business units; decisions take weeks and outcomes are rarely measured.



TARGET STATE
One control tower: correlated signals become owned decisions within days, with outcomes tracked against targets.



What We Built

- **Multi-domain view:** two heatmaps compare all business units and functions on one scale.
- **Agents & correlation:** 5 simulated agents feed a correlation engine; humans approve decisions.
- **Golden thread:** one cross-team scenario plays live on the simulation clock.
- **Programmes:** status by business unit and the not-on-track list.

Frameworks Applied With Metrics

Balanced Scorecard

Cynefin

RAPID

OODA loop

Named decider on every decision 30% → 100% (RAPID); average open-decision age 19 → 6.4 days, target 3.

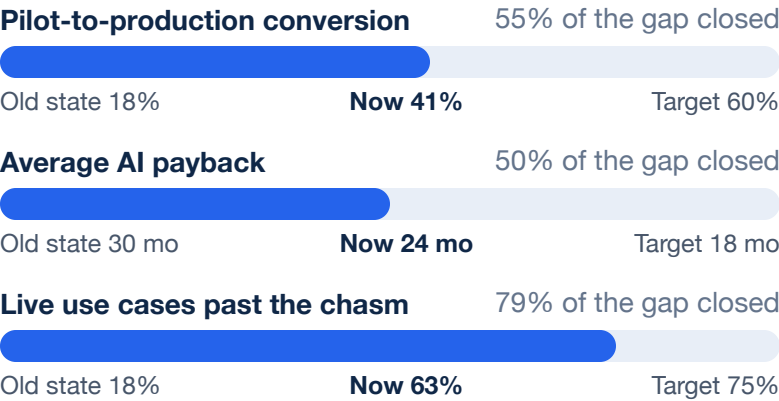
Suman: AI Transformation

FROM PILOTS TO MEASURABLE VALUE, FOR ANY ORGANISATION **BP1:** Identify ROI; transform any organisation to AI

OLD STATE
AI pilots everywhere, few in production; ROI guessed after the money is spent.



TARGET STATE
A governed AI portfolio where every use case has ROI identified before investment and pilots are promoted through gates.



What We Built

- **Readiness:** AI maturity radar per business unit.
- **Portfolio & gates:** 38 AI use cases, funnel, value versus feasibility, gate review with promote.
- **ROI & benefits:** NPV, payback and Monte Carlo P10 to P90 range.
- **Onboard any organisation:** 6 industry templates generate a plan and a Business Model Canvas.

Frameworks Applied With Metrics

Three Horizons

Diffusion of Innovations

Crossing the Chasm

Business Model Canvas

Stage-Gate

Value vs Effort

Three Horizons mix H1 38% / H2 40% / H3 22% versus a 70 / 20 / 10 guide shows "pilot purgatory": too little in live use cases.

Vaibhav: Regulatory & AI Governance

COMPLIANCE AT THE SPEED OF CHANGE; TRUSTWORTHY AI AND DATA **BP1:** Automate regulatory compliance (Banking); DPDP and AI governance

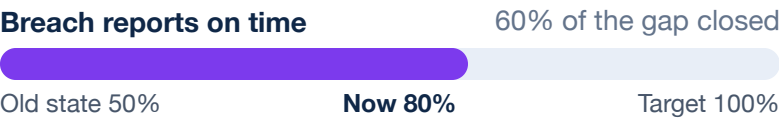
OLD STATE

Regulatory circulars mapped by hand over weeks; AI models and personal data not consistently governed.



TARGET STATE

Circular to obligation to control to evidence in days, every model registered and monitored, DPDP ready.



What We Built

■ Regulatory change:

a live circular arrives, obligations are mapped to controls and approved.

■ AI model register:

PSI drift monitoring; the fraud model at PSI 0.37 is restricted.

■ Data governance & DPDP:

consent coverage, lineage and a live breach clock.

■ Audit readiness:

evidence per control for internal audit and the regulator.

Frameworks Applied With Metrics

Three Lines Model

Risk process

Risk heat map

NIST AI RMF (spirit)

Three Lines applied to 33 high and critical risks: first line owns, second oversees, third assures.

Santhosh: Technology Spend & Investment

EVERY RUPEE TRACEABLE TO A SERVICE AND A VALUE **BP1:** Governance system for investment; CAPEX and OPEX

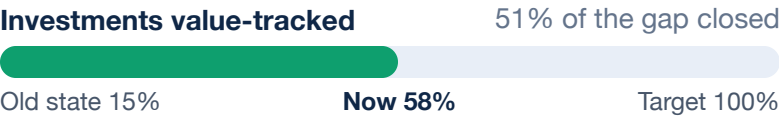
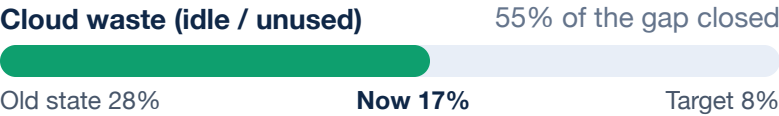
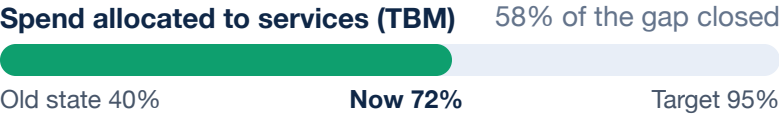
OLD STATE

Technology spend in cost pools nobody can trace; cloud waste; investments approved without benefit tracking.



TARGET STATE

Spend allocated to services, cloud under FinOps control, and every investment value-tracked after go-live.



What We Built

- CAPEX / OPEX & TBM: a Sankey from cost pools to towers to services.
- Budget & forecast: Holt-Winters forecast and EWMA anomalies, for example the festive cloud spike.
- FinOps & licences: rightsizing and licence utilisation.
- Investment governance: NPV, IRR, payback; security programmes judged on risk reduced.

Frameworks Applied With Metrics

Run-Grow-Transform

TCO

Sensitivity (tornado)

NPV / IRR

TBM

FinOps

Run-Grow-Transform gap from the 60 / 25 / 15 target mix reduced from 30 to 14.2 points; "Grow" is under-funded.

Pankaj: ERP RCA & Capacity

FIX CAUSES, NOT SYMPTOMS; CAPACITY BEFORE THE PEAK **BP1:** ERP root-cause analysis; infrastructure capacity; production automation

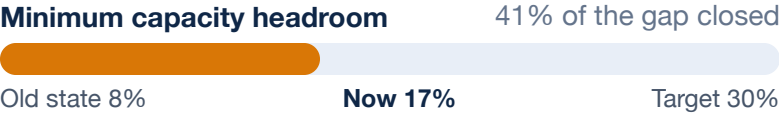
OLD STATE

The same ERP incidents recur; capacity is added after the peak hurts; production steps rely on manual hand-offs.



TARGET STATE

Root causes fixed permanently, capacity headroom ready before month-end and festive peaks, production steps automated.



What We Built

- ERP RCA:** Pareto of incident causes and a 5 Whys chain to the root cause.
- Capacity & forecasting:** Holt-Winters forecast of utilisation against the limit.
- Peak scenarios:** festive slider; 0% extra capacity gives 1,546 ms, 25% brings latency within target.
- Production automation:** automated steps tracked with a Lewin change strip.

Frameworks Applied With Metrics

Theory of Constraints

Pareto 80/20

5 Whys

ITIL problem mgmt

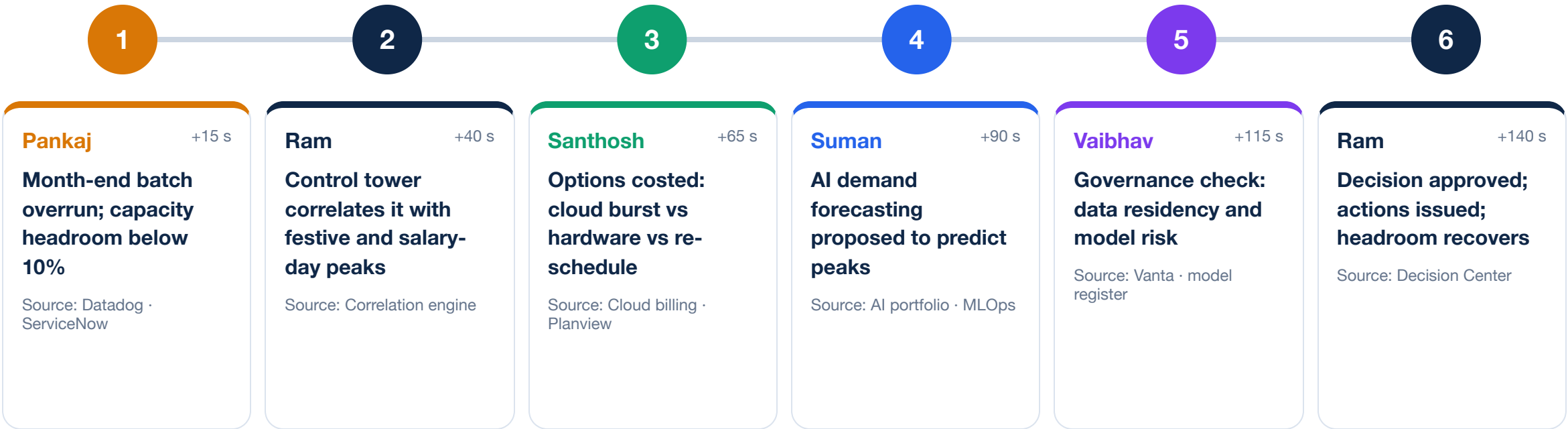
SRE error budget

Lewin

Theory of Constraints names the constraint per peak (for example month-end: IIoT ingestion CPU) and the next one after it.

The Golden Thread: One Connected Decision

Six steps play live on the simulation clock, from first signal to approved decision and recovered capacity.



- **Scenario: "Month-End Close Meets Festive Peak".** Every step cites real simulated records, so the audience can click through to the evidence.
- **The point:** no single owner or tool could solve this alone; CTO360 connects five owners into one decision with a measured outcome.

Frameworks Applied, Not Just Listed

"You do not need 100 frameworks." Choose the right one when a CEO, CFO or COO brings a problem, and measure it.

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Frameworks applied with tracked metrics

Balanced Scorecard, Cynefin, RAPID, OODA, Three Horizons, Diffusion, Chasm, BMC, Three Lines, Run-Grow-Transform, TCO, Sensitivity, Theory of Constraints, Lewin, Kotter, ADKAR

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Frameworks built into the screens earlier

Stage-Gate, Value vs Effort, NPV / IRR / ROI, maturity models, risk process and heat map, RACI, ITIL, 5 Whys, Pareto, SRE, TBM, FinOps, TEDIF

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Analytics techniques computed live

Holt-Winters, EWMA, Monte Carlo, NPV / IRR, S-curve, PSI drift, Pareto, queueing, error budget, glide path

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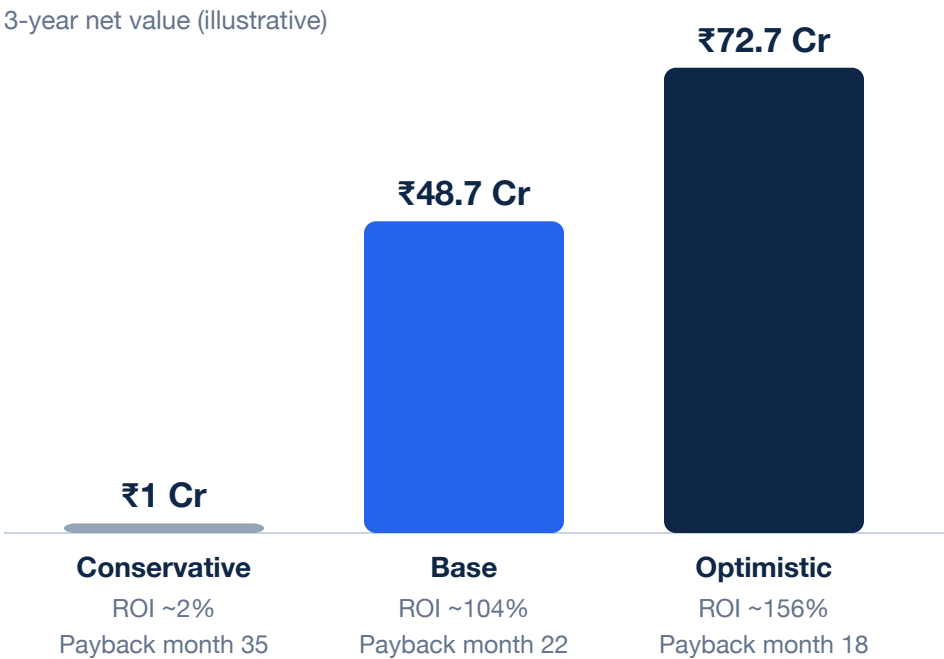
Reference library cards

Every framework, technique and concept explained with a small diagram and the business question it answers

- **Lean by design:** a framework is applied only where it serves an owner's business problem and produces a tracked metric; the rest stays in a reference library.
- **Team level change:** Kotter 8 steps turn green as decisions are approved in the app; ADKAR adoption score 1.8 → 3.2 out of 5 (target 4).

Value Case: Commit to the Conservative Case

A realistic, defensible value case: modest commitment, credible upside, and risk avoidance on top.



- **Investment:** about ₹30.7 Cr to build over 2 years, plus 10 to 20% a year to run.
- **Value ramp:** 15% of target value in year 1, 45% in year 2, 75% in year 3.
- **We commit to the conservative case** and show the base case as the expectation; payback around month 22 in the base case.
- **Risk avoidance is extra:** DPDP penalties for failed safeguards can reach about ₹250 Cr; one avoided incident can fund the programme.
- **Operational gains already modelled:** regulatory mapping 21 → 4 days, manufacturing RCA 38 → 9 hours, CAPEX approval 45 → 21 days.

Why CTO360 Is Different

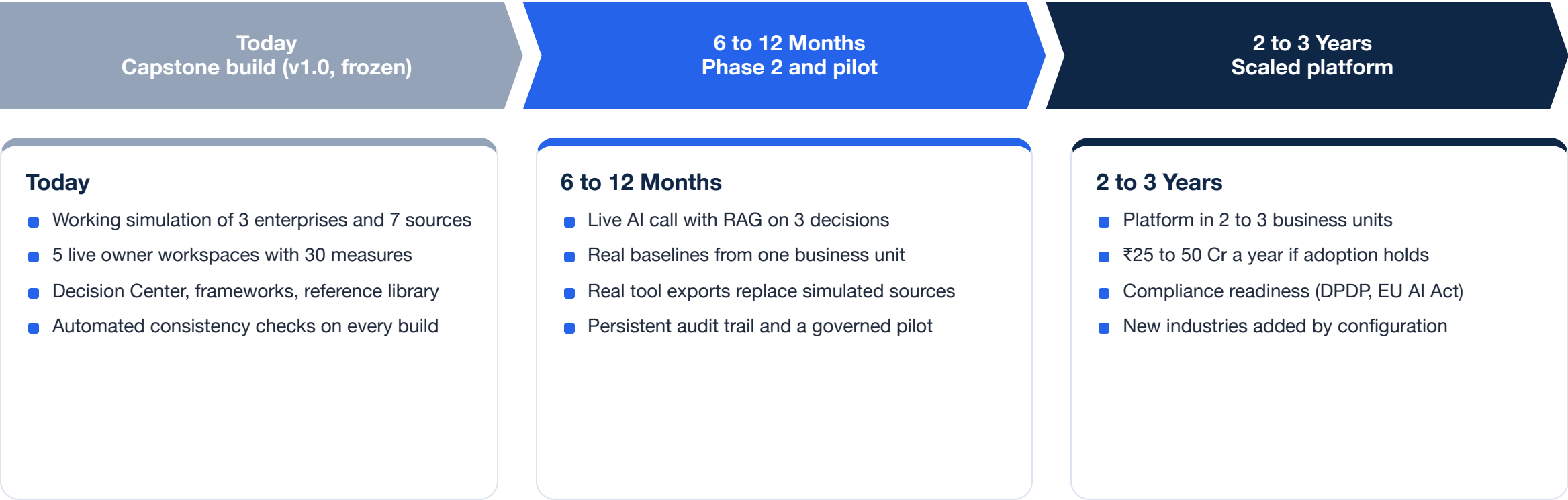
Point tools are essential, but each answers its own question. CTO360 answers the CTO's question: what matters most, and what should we decide?

Capability	Point tools (Planview, LeanIX, ServiceNow, Datadog, ...)	CTO360
Shows what is happening inside one function	✓ Yes	✓ Yes
Connects signals across tools into one story	✗ No	✓ Yes
Prioritised Top 10 decisions with owner and deadline	✗ No	✓ Yes
Tracks decision to action to measured outcome	🕒 Partial	✓ Yes
One comparable view across business units	✗ No	✓ Yes
Governed AI with named human decider and audit trail	🕒 Varies	✓ Yes
New industry added by configuration	✗ No	✓ Yes

■ We do not replace existing tools; we connect them. CTO360 is the decision layer a CTO needs on top of the tools the enterprise already pays for.

Roadmap: From Simulation to Production

The blueprint is ready. The next step is real data and live AI, under the same governance.



Guardrail from TEDIF: no production AI before day 90; every phase passes a human-approved gate.

IN ONE LINE

Five Owners. One Connected Decision. Measured Outcomes.

AI Recommends. Humans Decide.

Enterprise tools tell leaders what is happening inside each function. CTO360 connects those signals to show the CTO what matters, where to intervene, what decision to make, and whether it worked.

Thank you · Questions?

Live: cto360.vercel.app

The 16 Enterprise Functions

Every function is scored the same way in every business unit, which makes the heatmap comparable.

Technology Owner: CTO / CIO	Enterprise Architecture Owner: CTO / CIO	Engineering & R&D Owner: CTO	Data & AI Owner: CDO / CAIO
Cybersecurity Owner: CISO	Product & Innovation Owner: CPO / CTO	Corporate Strategy Owner: CEO / CSO	Finance Owner: CFO
Operations Owner: COO	Sales Owner: CRO	Marketing Owner: CMO	Customer Experience & Service Owner: CXO / COO
Human Resources Owner: CHRO	Risk & Compliance Owner: CRO	Legal Owner: CLO	Procurement & Vendor Management Owner: CPO / COO

Shaded: 6 technology functions owned by the CTO organisation. Unshaded: 10 business functions where technology shows up as a signal. Each function has 6 metrics: strategy, performance, cost, technology, risk and transformation.

The 7 Source Systems and the Question Each Answers

Each source is simulated with realistic records and linked to the others through shared record IDs.

Simulated source	Area	The question it answers
Planview	Strategy & Portfolio	Are investments aligned to strategy and on track?
SAP LeanIX	Enterprise Architecture	Is the application estate fit for the future?
Celonis + SAP Signavio	Process Intelligence	Where does the process lose time or money?
ServiceNow SPM	Portfolio & Workflow	Is demand flowing through to delivery?
Jellyfish	Engineering Intelligence	Is engineering capacity on the right work?
Datadog	Operations & Observability	Are critical services healthy and fast?
Vanta	Security, Risk & Governance	Are controls effective and risks within appetite?

Product names are used only to describe tool categories. CTO360 is an independent demonstration on synthetic data and is not affiliated with or endorsed by these vendors.

Anticipated Questions

Short, honest answers to the questions we expect from the panel.

Is this real data?

No. It is a seeded, realistic simulation; every score is calculated and cross-checked automatically. Real tool exports are the next step.

Why not just buy Planview or ServiceNow?

We do not replace them, we connect them. None correlates across tools into a prioritised decision with an owner and a measured outcome.

Where is the AI?

Simulated agents, correlation and recommendations today (rule-based and labelled). Next: a live LLM call with RAG on 3 decisions.

How do you stop AI from making bad decisions?

AI only recommends. A named decider approves at a gate (RAPID), models are monitored for drift (PSI), and the Three Lines Model provides oversight.

How does it work for a new industry?

Configuration, not rebuild. Retail was added by configuration only; the Onboard wizard creates a plan for any industry.

What does it cost and when does it pay back?

About ₹30.7 Cr build; base payback around month 22; we commit to the conservative case and treat risk avoidance as upside.

Disclaimer

Full disclaimer: cto360.vercel.app/about

- CTO360 is an independent capstone demonstration built entirely on **synthetic, simulated data**.
- All organisations (Meridian Bank, Arvant Industries, Orbit Retail), people, figures and events are illustrative.
- Third-party product names are referenced solely to describe tool categories and conceptual integration scenarios. CTO360 is **not affiliated with or endorsed by** the referenced vendors.
- Financial figures, ROI and payback are modelled estimates for discussion, not forecasts or advice.
- Framework applications are simplified for teaching purposes; they are not certifications or audits.